Day7
String Methods in Python

SKILL COURSE

String Methods in Python

Cleaning • Formatting • Validation • Transformation

SKILL COURSE

What Are String Methods?

TOOLS TO CLEAN AND MODIFY TEXT

USED IN DATA CLEANING & FORMATTING

ESSENTIAL FOR DATA ANALYSTS

Why String Methods Matter?

Clean	Standardize	Fix	Remove	Validate	Extract
Clean customer names	Standardize city/state names	Fix letter casing	Remove unwanted spaces	Validate emails & phone numbers	Extract meaningful text

strip(), upper(), lower()

- strip(): Removes extra spaces
- upper(): Converts text to UPPERCASE
- lower(): Converts text to lowercase

- Use Case:
- Clean names, cities, product titles

SKILL COURSE

title(), capitalize(), replace()

- title(): First letter caps in each word
- capitalize(): First letter only caps
- replace(): Replace wrong spellings or characters

- Use Case:
- Fix messy text from CSV files
- Standardize product names

SKILL COURSE

split() & join()

- split(): Break text into words
- join(): Combine list items into a string

- Use Case:
- Extract first/last name
- Build clean labels or IDs

find(), index(), startswith(), endswith()

- find(): Returns position of substring
- index(): Same but gives error if not found
- startswith(): Checks beginning text
- endswith(): Checks ending text

- Use Case:
- Extract domain from email
- Validate file formats (.csv, .xlsx)

isnumeric(), isalpha(), isalnum(), count()

- isnumeric(): Numbers only
- isalpha(): Alphabets only
- isalnum(): Alphabets + numbers
- count(): Count letters/words

- Use Case:
- Validate phone numbers & product codes

Cleaning Special Characters

- Use replace() to remove:
- ₹, %, , /, : , etc.

- Example:
- 'Price: ₹4500/' → '4500'

Practical Use Cases

- Clean names & cities
- Fix spelling issues
- Extract domain/year
- Validate inputs
- Remove unwanted characters

Day 7 Practice Tasks

1. Clean names
2. Fix city names
3. Extract email domains
4. Correct spellings
5. Remove symbols from price fields

Practice Code

```
text = "   hello python learners 123   "

print("Original:", text)

# Try different methods
print("Upper:", text.upper())
print("Lower:", text.lower())
print("Title:", text.title())
print("Strip:", text.strip())
print("Split:", text.split())
print("Replace:", text.replace("python", "Java"))
print("Find:", text.find("python"))
print("Count:", text.count("e"))
print("Starts With:", text.startswith("hello"))
print("Ends With:", text.endswith("123"))
print("Is Digit:", "123".isdigit())
print("Is Alpha:", "python".isalpha())
print("Is Alnum:", "python123".isalnum())
print("Swapcase:", "Hello PYTHON".swapcase())
print("Zfill:", "50".zfill(6))
```

Final Takeaway

- String methods are the backbone of text cleaning.
- Master them for realworld data analysis.